When mIoU Lies: Topology-Aware Evaluation of Thin Road-Marking Segmentation

Anonymous Submission

Abstract

Mean Intersection-over-Union (mIoU) is the de-facto metric for semantic segmentation, including road-marking perception. We show two ways it fails for *thin* (2–8 px) road markings. First, on **converged** models in a width-controlled study (3 seeds, 95% CIs), mIoU under-reports correctly and continuously detected thin markings: at 2 px, mIoU $= 0.35 \pm 0.01$ while centerline-Dice $= 0.997 \pm 0.001$ and tolerance-IoU $= 0.98$—a $\sim$0.65 gap with non-overlapping confidence intervals. Second, under extreme multi-class imbalance (real RLMD, 25 classes, 609:1), mIoU—in *both* its all-class and foreground forms—*mis-ranks* losses: its top-ranked model is a frame-flooder (0.9% pixel accuracy) and it places the genuine best detector third. Frequency-weighted IoU and pixel accuracy avoid rewarding flooding but are insensitive to foreground quality (spread ≤ 0.005 vs. centerline-Dice's 0.31); only the topology metric is both flood-robust and sensitive. We propose a thin-structure evaluation protocol and an automated three-test check, and recommend reporting it alongside mIoU. A studied connectivity loss yields no reliable multi-seed gain and is supplementary. Results are on ground-view markings and a controlled synthetic benchmark; aerial and full-scale validation are future work.

1 Introduction

Road-marking segmentation underpins lane perception, HD-map maintenance, and infrastructure auditing. Markings are thin and sparse: in our real dataset they occupy under 2% of pixels, with a 609:1 imbalance between the most- and least-common classes. The field reports mIoU almost universally. We ask whether mIoU faithfully reflects thin-marking quality and selects the right model and loss. It does not. Our contributions are: (1) a thin-structure evaluation protocol (clDice, Boundary-IoU, tolerance-IoU, width-stratified analysis); (2) empirical evidence that mIoU mis-ranks losses on real markings, crowning a model that floods 99% of pixels, with the mechanism measured directly via pixel accuracy; (3) an honest multi-seed study of a connectivity loss that yields no reliable gain; and (4) a reproducible benchmark with a deployment path. This is an *evaluation* paper: we claim no new state of the art and report a loss that does not help.

2 Related Work

U-Net, U-Net++, DeepLabV3+, PSPNet, and SegFormer span the architecture space. clDice with its differentiable soft-skeleton, and Boundary-IoU, were proposed for tubular structures and boundary quality; we adopt them and add a width-stratified view. Dice, Focal, and Tversky address imbalance; that overlap/accuracy metrics reward over-prediction is known. Our contribution is the quantified rank-inversion on real road markings, not the observation that imbalance is hard.

3 Methodology

Protocol. For binary any-marking masks we report clDice (harmonic mean of skeleton precision/sensitivity), Boundary-IoU (IoU on a d-px boundary band), tolerance-IoU (detection within τ px of the GT centerline), and a width-stratified IoU/clDice where GT marking width is an independent variable. **Loss.** We study $\mathcal{L} = w_{ce}\mathrm{CE} + w_r\mathrm{Dice} + w_t(1 - \text{soft-clDice})$. **Data.** RLMD (2,137 real images, 25 marking classes) is the real benchmark; a controlled synthetic generator provides ground-truth width control.

4 Experimental Setup

Single 4 GB GPU; PyTorch; fp32. U-Net++ (ResNet-34) for the loss ablation. ImageNet-pretrained encoders, AdamW, cosine. Compute-bounded: RLMD runs use a 500-image subset, 12 epochs, 384^2. Statistics: per-image bootstrap 95% CIs, paired bootstrap tests, seeds 42/123/999. We disclose that the real 25-class models are under-trained (best clDice 0.37).

5 Results

5.1 mIoU under-reports thin markings (converged, multi-seed)

On converged models (clDice ≈ 0.997 at the training width), three seeds, evaluated across the controlled width series (Table **??**; mean $\pm$ 95% CI): at 2 px mIoU $= 0.35 \pm 0.01$ while clDice $= 0.997 \pm 0.001$ and tolerance-IoU $= 0.98$. The lines are correctly and continuously detected; the $\sim$0.65 gap has non-overlapping, tight CIs and is *not* an under-training artifact. This is the paper's primary, hardest-to-attack result.

Table 1: Width study, converged models, mean $\pm$ 95% CI over 3 seeds. mIoU collapses on thin markings; topology/tolerance metrics do not.

width	mIoU	TolIoU	clDice
2 px	0.35 ± 0.01	0.98 ± 0.01	0.997 ± 0.001
4 px	0.89 ± 0.05	1.00 ± 0.00	0.998 ± 0.002
8 px	0.47 ± 0.06	0.97 ± 0.01	0.95 ± 0.05

Table 2: RLMD loss ablation (U-Net++, same architecture). Tversky/Dice top mIoU but flood the frame (PixelAcc ≈ 0.008).

Loss	mIoU	PixAcc	clDice	clPrec	clRec
CE+Dice	0.029	0.983	**0.373**	0.907	0.356
CoRe	0.005	0.981	0.275	0.896	0.211
Focal	0.018	0.983	0.085	–	0.103
CE	0.010	0.982	0.063	–	0.096
Dice	0.140	0.008	0.062	–	1.000
Tversky	**0.144**	**0.009**	0.084	0.133	1.000

5.2 mIoU mis-ranks losses (real RLMD)

Table **??** gives the same-architecture loss ablation. We test every common metric (Table **??**): does its top model flood the frame; does it rank the genuine best detector first; is it foreground-sensitive? mIoU—in both its all-class and foreground forms—ranks a frame-flooder first and the best detector third; FWIoU and pixel accuracy avoid flooding but are insensitive (spread $\leq$ 0.005 vs. clDice's 0.31); only clDice passes all three. We scope this as a multi-class extreme-imbalance phenomenon, complementing Sec. V-A.

5.3 Connectivity loss: honest multi-seed result

At seed 42, CE+Dice beats CoRe (clDice 0.37 vs. 0.28, $p < 0.001$). Across three seeds the advantage vanishes: CE+Dice 0.405 ± 0.058 vs. CoRe 0.406 ± 0.141 ($\Delta = -0.001$, 95% CI $[-0.18, 0.18]$; CoRe wins at seed 999). The single-seed result was an artifact caught only by multi-seed evaluation.

5.4 Deployment

ONNX Runtime is $1.47\times$ faster than eager PyTorch on CPU at bit-identical output (max $|\Delta| = 9\text{e}{-8}$); a C++/LibTorch engine and TensorRT path are released.

6 Discussion and Limitations

For thin road markings, report clDice/Boundary-IoU/tolerance-IoU and pixel accuracy, not mIoU alone: mIoU can rank a useless frame-flooding model first. **Limitations.** (1) RLMD is ground-view; we report no real aerial/UAV marking results. (2) The multi-class real models are under-trained (clDice 0.37), but the *primary* result (Sec. V-A) uses converged models (clDice $\approx$ 0.997, 3 seeds, non-overlapping CIs) and a converged binary-marking model on real RLMD plateaus at clDice 0.64 / mIoU 0.53, $so the central claim does not rest on under-training. (3) The real-mis-ranking is scoped to multi-class extreme imbalance. (4) Three seeds; the cross-dataset probe is confounded; the studied loss gives no reliable gain.$

Table 3: Three-test metric check on RLMD (extreme imbalance). Only clDice passes all three.

Metric	top floods?	best #1?	spread
mIoU (fg)	yes	no (#3)	0.023
mIoU (all)	yes	no (#3)	0.023
FWIoU	no	yes	0.005
PixelAcc	no	yes	0.003
clDice	no	yes	**0.310**

7 Conclusion

mIoU is an unreliable—sometimes rank-inverting—metric for thin road-marking segmentation. We provide a topology-aware protocol that ranks models correctly, a controlled demonstration of the mechanism, an honest multi-seed loss study, and a reproducible benchmark. Converged real-data and aerial/UAV validation are the key next steps.

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